

**BRIDGING REALITIES: EFFECTIVENESS OF AI AND
IMMERSIVE TECHNOLOGIES IN CROSS- PLATFORM
NEXT-GEN MUSEUM EXPERIENCES FOR
STORYTELLING AND INTERACTION**

Project ID: 25-26J-416

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INTERACTIVE AR EXPLORATION OF EARTH'S GEOLOGICAL HISTORY

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DECLARATION

I declare that this is my own work and this dissertation does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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DEDICATION

This dissertation is dedicated with sincere gratitude to my supervisor, Mr. Aruna Ishara Gamage, and my co-supervisor, Mr. Nushkan Nizme, for their continuous guidance, support, and encouragement throughout the research process. Their valuable advice and constructive feedback greatly helped me to complete this work successfully.

I also dedicate this work to my family and friends, who supported me with patience, motivation, and understanding during every stage of this journey. Their encouragement gave me the strength to continue working hard and overcome the challenges faced during the project.

Finally, this work is dedicated to everyone who contributed directly or indirectly to the successful completion of this research project.

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ABSTRACT

Traditional museum displays often find it difficult to explain Earth's geological history because fossils, rocks, minerals, and geological era exhibits are usually presented through static objects and short descriptions. This research develops an interactive mobile Augmented Reality learning system for the Colombo National Museum's geology gallery to transform passive viewing into story-driven, participatory learning. Using QR, WebAR and image-marker scanning, visitors can access Augmented Reality experiences for the Paleozoic, Mesozoic, and Cenozoic eras, supported by 3D visualizations, animations, on-screen information panels, and AI-supported narration. An AI guide and chatbot provide contextual explanations, answer visitor questions, and support personalized learning. The system also includes a digital Coloring to Augmented Reality activity, allowing users to color prehistoric illustrations and view them as animated Augmented Reality outputs. Learning impact is evaluated through era based pre- and post-experience quizzes, usability feedback, and visitor interaction observations. The study offers a low-cost approach for improving engagement, curiosity, and understanding of Earth's natural history within museum learning environments.

Keywords: Augmented Reality, Mobile AR, Museum Learning, Geological Eras, QR-Triggered AR, AI Guide, AI-Supported Narration, Color-to-AR, Interactive Education, 3D Visualization, Earth's Geological History

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List of Abbreviations

Abbreviations	Descriptions
AR	Augmented Reality
VR	Virtual Reality
XR	Extended Reality
MR	Mixed Reality
AI	Artificial Intelligence
QR	Quick Response (Code)
3D	Three-Dimensional
SUS	System Usability Scale
RFID	Radio Frequency Identification
SDK	Software Development Kit
API	Application Programming Interface
UNESCO	United Nations Educational, Scientific and Cultural Organization
STEM	Science, Technology, Engineering and Mathematics
WebAR	Web-based Augmented Reality
WebXR	Web Extended Reality

1. INTRODUCTION

1.1 Background Study

Sri Lankan museums safeguard a rich record of cultural and natural heritage, yet presentation styles still lean on static display cases, labels, and brief captions. This makes it difficult for visitors to build a clear storyline or connect local heritage to wider world perspectives. Passive media, and low interactivity fall short of the expectations of today's younger generation, as well as tourists and international guests who are accustomed to hands-on, digital experiences. As a result, the educational value of a visit is often under-realized, and many visitors leave without fully appreciating the richness and significance of the collections.

At the Colombo National Museum complex, the Natural History wing illustrates this challenge. The geology gallery presents important fossils, rocks, and minerals, but the experience is largely text-led and object-centered. Geological ideas-plate movement, fossilization, mass extinctions, and climate shifts-span vast timescales and remain abstract when encountered only through specimens and captions. Younger visitors, in particular, struggle to link what they see with the processes and stories that shaped Earth over millions of years.

To make the experience more engaging without disrupting the integrity of the collection, the study layers light-touch Augmented Reality onto existing displays through mobile AR and WebAR access. Short scan-to-learn moments near wall art and exhibit points situate each geological era with concise narratives and animated visuals, while a simple Color-to-AR activity lets children bring their scenes to life on a phone. The QR scan workflow is important because it allows visitors to open selected content quickly through a browser-based WebAR route while the mobile AR component supports richer image-target-based interaction. Complementary pieces across the wider gallery, such as a guided VR fossil-dig, an AI-supported mobile guide, and a family-friendly wilderness experience, support different age groups and language needs, helping visitors follow what they are seeing at the point of interest.

By bringing these strands together, the work shifts geology from passive viewing to guided participation. Anticipated outcomes include higher engagement among children and families, stronger recall of deep-time concepts, clearer pathways for multilingual and mixed-ability audiences, and a more coherent visit that links objects, processes, and place delivered with minimal changes to the gallery. The result is a scalable approach to modernizing the museum experience while preserving curatorial integrity and scientific accuracy.

The difficulty is not that the exhibits lack educational value, but that the knowledge embedded in them is often hidden behind scientific terminology and long geological timescales. A fossil, a rock sample, or a mineral specimen can represent millions of years of environmental change, but a visitor usually sees only the final preserved object. Without additional interpretation, the connection between the object and the process that formed it can remain unclear. This is particularly important for geological learning because the subject depends on understanding sequence, transformation, scale, and cause-and-effect relationships rather than only identifying isolated objects.

The geology gallery therefore requires a learning approach that can connect physical specimens with visual explanation. For example, the Paleozoic era can be explained more clearly when visitors see early marine life and changing land environments; the Mesozoic era becomes more memorable when prehistoric species and climate conditions are visualized; and the Cenozoic era is easier to understand when visitors can connect more recent mammals and environmental changes with present-day biodiversity. These examples show why an era-based storytelling method is suitable for the museum context: it helps visitors move from a single exhibit toward a connected timeline of Earth history.

Mobile AR is appropriate for this purpose because they add digital interpretation without replacing the real museum object. Visitors can remain in front of the actual exhibit while viewing supplementary 3D scenes, short narration, labels, and interaction prompts. In the QR scan workflow, WebAR reduces the barrier to entry because selected AR content can be opened through the browser, while the app-based AR workflow can support more controlled image-target recognition, scene management, and guided interaction.

This balance is suitable for a museum environment where visitors may have different devices, time limits, and levels of technical familiarity.

The Bridging Realities project uses this idea as part of a broader cross-platform museum learning framework. The individual AR component developed in this study focuses specifically on geological history and uses QR-triggered WebAR access, app-based AR visualization, AI-supported narration, Color-to-AR interaction, and quiz-based evaluation to make the learning flow more active, accessible, and measurable.

1.2 Literature Survey

Augmented reality (AR) has become a practical way to enhance museum and educational experiences by overlaying 3D models, multimedia, and interaction on physical artifacts and printed media. Several recent systems demonstrate how AR helps museums expose hidden context and make exhibits interactive for remote or on-site visitors; these works typically build mobile apps that present 3D reconstructions, audio guides, or media layers that visitors can explore with phones or tablets [1], [2], [3].

A second cluster of papers focuses on coloring books and children's learning. Live texturing and coloring-book AR systems convert a child-colored page into animated 3D characters or scenes, often using fast texture mapping or 2D-to-3D model pipelines to keep interaction immediate and playful [4], [5], [6]. These works show that creative input can improve engagement, but they also highlight technical challenges such as color capture, marker stability, lighting variation, and lightweight rendering on mobile devices.

A related strand covers general AR learning platforms and interactive textbooks. Projects such as ARLearn and Learnscan show how simple markers or QR codes can connect learners to AR viewers, 3D models, multilingual voiceover, and classroom deployment workflows [7], [8]. Marker-based AR research also shows that low-cost scanning mechanisms can support accessibility when the experience is designed for ordinary mobile devices rather than specialized equipment [9].

Museum-scale AR studies provide useful implementation lessons. AR Museum and Interactive Virtual Museum systems demonstrate virtual navigation, modular exhibit rooms, and 3D artifact inspection [1], [2]. MARHN-style natural history museum systems show the value of combining 3D models and video-based interpretation for museum visitors [3]. Site-specific cultural heritage deployments such as the Wariwilca Museum AR application and SMB II Museum study show positive learning and satisfaction outcomes, although their evaluations remain limited in scale and transferability [10], [11].

Across these areas, method and evaluation choices are consistent: most studies use small in-lab or single-site deployments such as classroom pilots, museum demos, or focus groups to measure usability, immediate engagement, and short-term learning gains. Papers that report quantitative learning data typically use pre-test and post-test instruments, usability metrics, or visitor satisfaction measures. These studies are useful, but they often emphasize immediate motivational outcomes rather than long-term retention, repeated use, or transfer across different museum contexts.

On the implementation side, several contributions detail content-creation pipelines, including marker design, 3D modeling, UV mapping, texture budgets, polycount constraints, and reusable asset specifications. The mobile coloring work describes model and texture specifications intended to keep rendering stable on midrange devices, while live-texturing papers show how color capture can be connected to texture mapping in real time [4], [5], [6]. These pipelines are directly relevant to this project because the proposed component also combines 3D geological assets, narration, QR or image markers, and digital coloring content.

Several works address accessibility, deployment constraints, and practical affordances. Field and school deployments describe limitations such as device battery life, camera quality, screen size, holding fatigue, lighting variation, and connectivity requirements [10], [12]. These constraints are important for the present study because museum visitors may use different devices and may not want to install a full application for a short interaction. For this reason, the QR scan workflow also uses WebAR so that selected scenes can be accessed through a browser-based route supported by WebXR and web-based 3D model delivery [13], [14].

There is beginning work on personalization and adaptive presentation, but it is limited. A few museum systems include simple personalization, such as audio tracks or gesture-based content switching, while child-centered systems such as color-sensing AR and dynamic visualization applications explore imagination, play, and interaction for younger learners [15], [16]. However, concrete adaptive storytelling systems that adjust to age, prior knowledge, or measured interest are still more often proposed as future work than implemented and evaluated in museum contexts.

Finally, a technical theme that recurs is robustness. Marker detection under varied real-world coloring, domain gaps between clean scans and museum photos, sensor noise, browser compatibility, and changing lighting conditions can all reduce the reliability of AR learning experiences [5], [6], [13], [14]. Likewise, the Learnscan and ARLearn projects call out lighting, occlusion, and camera calibration as ongoing challenges in classroom scenarios [7], [8]. These concerns inform the design of this study, where WebAR QR access and app-based image target recognition are treated as complementary rather than competing routes.

The reviewed museum AR literature shows that visualization alone is not sufficient for a complete educational experience. Systems such as AR Museum and Interactive Virtual Museum show that AR can attract attention and support artifact inspection, but a museum learning system also needs a coherent story, clear instructions, meaningful interaction, and evaluation evidence [1], [2]. This is especially important for geological history because visitors must connect separate concepts such as fossilization, extinction, climate change, and evolutionary development.

Natural history museum AR applications also indicate the importance of connecting digital content to physical context. MARHN-style mobile AR systems show that 3D models and multimedia can improve exhibit interpretation [3]. However, the present component extends this idea by connecting QR-triggered WebAR access, app-based AR scenes, AI-supported narration, Color-to-AR interaction, and quizzes into one learning sequence rather than treating each feature as a separate attraction.

The coloring and live-texturing literature is highly relevant to the child-focused part of this component. Live texturing systems demonstrate how a user's colors can be transferred to a digital character, while mobile AR coloring books show that creative interaction can support engagement among children [4], [5], [6]. This project applies the same principle to geological learning by linking coloring activities to prehistoric life and era-based storytelling rather than using coloring only as entertainment.

Marker-based, QR-supported, and WebAR learning systems provide another useful foundation. ARLearn and Learnscan show that simple scanning mechanisms can connect physical materials to digital learning content [7], [8]. WebXR and web-based 3D model delivery further support the idea that selected AR content can be launched through the browser after a QR scan, reducing installation friction for short museum visits [13], [14].

A further theme in the reviewed studies is the role of evaluation. Several museum AR studies report improved satisfaction and engagement, but many rely on small samples, short interaction windows, or post-use feedback only [10], [11]. This study therefore includes pre-test and post-test quizzes, usability feedback, and interaction observations so that the AR and WebAR component is evaluated as a learning tool rather than only as a visual demonstration. Small local history museum information systems also show that AR can support artifact digitization and mobile interpretation through structured content pipelines [12]. However, these systems generally focus more on digitization and access than on narrative continuity. In contrast, this study organizes AR content around the Paleozoic, Mesozoic, and Cenozoic eras so visitors can follow a chronological geological story.

Studies on child-centered AR highlight the need to balance imagination and instruction. Color-sensing and dynamic visualization systems show that children can become more engaged when they contribute creatively to the digital output [15], [16]. However, for a museum context, the interaction must remain tied to scientific meaning. Therefore, the proposed Color-to-AR activity is connected to geological era content and supported by narration and quiz-based reflection.

Overall, the literature supports the use of AR, WebAR, and QR-based access for museum learning, but it also shows that these systems must be designed carefully to avoid becoming only short-term visual attractions. The present study addresses this by integrating exhibit-linked scanning, browser-based WebAR access, app-based AR visualization, storytelling, child-friendly creative interaction, AI-supported narration, and learning evaluation within the Bridging Realities framework.

1.3 Research Gap

1.3.1 Overall research project gap:

Existing museum AR/VR systems are often fragmented and presentation-focused, with limited integration of AI guidance, gamification, collaboration, accessibility, and learning analytics. This creates a need for an AI-assisted immersive museum platform that transforms static natural history exhibits into interactive, personalized, and measurable learning experiences.

1.3.2 Individual research component gap:

Current AR museum applications improve visual engagement, but many lack a complete learning flow with geological storytelling, AI guide/chatbot support, creative interaction, and learning assessment. This creates a gap in helping visitors understand Earth's geological history through interactive and measurable AR learning.

1.3.3 Research component gap analysis:

Although several studies have introduced augmented reality into museum settings with promising results, closer examination reveals limitations that restrict their broader educational and operational value. Marker-less museum applications such as AR Museum [1] show that virtual portals and navigation can improve visitor engagement, but their evaluations remain narrow in scope, focusing on immediate usability rather than sustained learning. Similarly, the Interactive Virtual Museum [2] demonstrates technical feasibility and media enrichment, yet it remains largely object-centric, with limited attention to narrative-driven geological storytelling or structured assessment.

Site-specific projects like the Wariwilca deployment [10] and the SMB II Museum study [11] confirm that AR can enhance learning and satisfaction, but these works are small-scale, short-term, and confined to individual institutions, limiting their generalizability. The Local History Museum information system [12] and MARHN architecture [3] provide useful pipelines for digitization, 3D content, and mobile integration, but they emphasize technical workflows more than reusable educational flow, WebAR accessibility, or quiz-based measurement of learning outcomes.

Taken together, these studies validate the potential of AR to make exhibits more interactive and accessible, yet they expose common gaps: evaluations are often restricted to short-term pilots, many applications remain object-centric, browser-based WebAR access is not always integrated with museum QR workflows, and narrative or child-friendly learning design is often underdeveloped. Addressing these shortcomings requires a more comprehensive approach that combines story-first geological experiences, mobile and WebAR access, sustainable content pipelines, and measurable evaluation.

AR Museum (Markerless Mobile Virtual Museum): This system proposes a marker-less AR museum application using portals to navigate virtual galleries. Visitors preferred AR navigation and reported higher usability. However, the study mainly tested the finished prototype with limited users and did not deeply measure learning retention, structured storytelling, or QR-to-WebAR accessibility for short museum interactions [1].

Interactive Virtual Museum (Unity + ARCore): This system developed a Unity/ARCore-based virtual museum offering modular rooms and 3D artifact inspection. It enables remote access and media-rich exhibits, but its focus remains on implementation and artifact viewing. It does not fully evaluate narrative impact, geological era learning, Color-to-AR interaction, or quiz-based learning outcomes [2].

Wariwilca Museum AR App (Cultural Heritage Interaction): This study deployed a mobile AR application in a heritage site and reported positive engagement and cultural learning outcomes. However, the evaluation was small-scale and specific to one museum, and it does not provide a reusable model for QR-triggered WebAR access, geological storytelling, or child-focused creative interaction [10].

AR Information System for a Small Local History Museum: This work introduced an AR information system that uses photogrammetry for artifact digitization and richer visitor interaction. While useful for documenting workflows, it emphasizes technical digitization and mobile presentation rather than learning assessment, WebAR-based access, or scalable authoring for era-based narratives [12].

SMB II Museum AR Learning Experience: This study implemented AR overlays for cultural artifacts and reported improved learning and visitor satisfaction. Still, the evaluation was short-term and focused mainly on factual overlays. It does not strongly address how narration, QR scanning, WebAR access, creative interaction, and quiz feedback can be combined into one visitor flow [11].

MARHN (Mobile AR for Natural History Museum: 3D + Video): This work presents an architecture and mobile application that integrates 3D models and videos for museum interpretation. The paper demonstrates engineering feasibility for natural history museum content, but it does not provide a full framework for WebAR QR access, Color-to-AR learning, AI-supported narration, or measurable geological era assessment [3].

Most existing museum AR studies show clear potential for improving visitor engagement, but they stop short of addressing narrative-driven design, WebAR accessibility, long-term or structured evaluation, and scalable content authoring within one system. Many systems are developed as isolated prototypes that focus on displaying 3D objects or improving visual interaction, while giving less attention to how visitors follow a complete learning journey from exhibit discovery to understanding and reflection. In addition, limited support for browser-based access can reduce accessibility for visitors who may not want to install a separate mobile application. Without these considerations, AR remains confined to small-scale demonstrations with limited transferability across different museum environments. Therefore, a more comprehensive approach is needed, one that emphasizes story-first experiences, mobile and browser-based access, sustainable content pipelines, and measurable learning outcomes to ensure reliability, usability, and educational value.

Table 1 summarizes how the proposed component responds to these gaps by combining user interaction, cultural and educational outcomes, narrative storytelling, skill-based engagement, WebAR-supported QR access, and evaluation depth within a single AR learning workflow.

Research Paper	User Interaction and Engagement	Cultural / Educational Outcomes	Narrative and Storytelling	Skill-Based Engagement	Evaluation Depth
AR Museum (Marker less Mobile Virtual Museum)	✓	✗	✗	✗	✗
Interactive Virtual Museum (Unity + ARCore)	✓	✗	✗	✗	✗
Wariwilca Museum AR App (Cultural Heritage Interaction)	✓	✓	✗	✗	✗
AR Information System for a Small Local History Museum	✓	✓	✗	✗	✗
SMB II Museum AR Learning Experience	✓	✓	✗	✗	✗
MARHN (Mobile AR for Natural History Museum: 3D + Video)	✓	✗	✗	✗	✗
Our Project (Mobile AR + WebAR)	✓	✓	✓	✓	✓

Table 1: Research Gap

When the existing systems are compared more closely, the gap becomes clearer. The AR Museum system contributes to navigation and virtual gallery interaction, but it does not deeply explore how visitors learn a connected scientific story from the experience [1]. In contrast, the proposed component uses scanning as an entry point into a guided geological story rather than as a navigation feature only.

The Interactive Virtual Museum system demonstrates the technical feasibility of building 3D exhibit spaces using Unity and ARCore [2]. However, its focus is mainly on presenting and inspecting museum artifacts. This leaves a gap in narrative design and assessment. The proposed system addresses this by combining era-based AR scenes, narration, Color-to-AR interaction, and quiz-based evaluation.

The Wariwilca Museum AR application and the SMB II Museum AR experience provide examples of cultural learning and visitor satisfaction [10], [11]. Yet both remain tied to specific sites and short-term evaluations. Their findings are valuable, but they do not fully explain how a reusable AR and WebAR workflow can support geological learning across museum exhibits.

The Local History Museum information system and MARHN architecture contribute important technical knowledge about museum digitization, photogrammetry, and mobile AR delivery [12], [3]. However, these contributions focus more on implementation pipelines than on a complete learning model. This study extends that direction by including QR-triggered WebAR access, guided narration, and learning assessment.

The coloring-based AR systems provide strong evidence for child engagement, but they also reveal another gap. Live texturing and mobile AR coloring systems successfully connect physical or digital coloring with animated output [4], [5], [6]. However, they are usually not designed around museum-specific geological content, era-based storytelling, or measurable visitor learning outcomes.

Therefore, the key gap addressed by this study is the absence of a compact and integrated AR museum learning component that combines era-based geological storytelling, QR-triggered WebAR access, app-based AR visualization, child-friendly Color-to-AR interaction, AI-supported narration, and quiz-based learning assessment in one workflow. Existing works address parts of this problem separately, but they do not provide a complete model for accessible geological learning in a Sri Lankan museum context.

1.4 Research Problem

1.4.1 Overall Research Problem

Traditional museum experiences still rely heavily on static exhibits and text-based explanations, which limit visitor interaction, engagement, accessibility, and learning retention, especially for younger and diverse audiences. Although AR and VR have been used in museum education, existing systems often focus on presenting information rather than supporting interactive learning, AI-assisted guidance, collaboration, and measurable learning outcomes. Therefore, the overall research problem is:

How can an AI-assisted immersive museum platform that combines AR, VR, gamification, collaboration, accessibility, and learning analytics transform static natural history exhibits into interactive, personalized, and measurable learning experiences for museum visitors?

1.4.2 Individual Research Problem

The geology gallery presents fossils, rocks, minerals, and geological eras through mostly static displays, making concepts such as fossilization, plate movement, mass extinction, climate change, and prehistoric life difficult for visitors to understand. Existing AR museum systems support visualization, but many lack a complete learning flow that combines geological storytelling, AI guide/chatbot support, creative interaction, and structured learning assessment. Therefore, the individual research problem is:

How can a mobile AR learning experience that combines QR/image-marker triggered geological era storytelling, AI guide/chatbot support, digital Color-to-AR interaction, and quiz-based assessment improve visitor engagement and understanding of Earth's geological history in a museum environment?

1.4.3 Individual Research Problem Overview

Museums around the world are entering a digital transformation era, where conventional display methods such as static artifacts behind glass and text-heavy captions are being challenged by the expectations of younger, digitally immersed audiences. In Sri Lanka, the Colombo National Museum's Natural History section exemplifies this challenge: vast geological concepts such as the Paleozoic, Mesozoic, and Cenozoic eras are presented through fossils and minerals, yet without enough interaction or narrative support. For many visitors, particularly students, these displays remain abstract and fragmented, leading to missed opportunities for deeper cultural and scientific learning.

The rapid growth of Augmented Reality and WebAR technologies has provided museums with practical ways to overlay interactive, contextualized, and animated content on existing exhibits. However, while AR applications can improve engagement, questions remain regarding usability, educational validity, accessibility, and sustainability in resource-limited contexts. Systems that rely only on high-cost hardware or full application installation can be impractical for short museum visits, while WebAR can reduce the access barrier by allowing selected content to open after a QR scan through the browser.

This research study investigates how AR storytelling, QR access, and Color-to-AR interaction can transform static museum exhibits into immersive learning environments that not only engage visitors but also support knowledge retention about deep-time geological eras. While AR-based cultural heritage projects have shown promise in enhancing visitor experience, several critical issues remain:

- **Narrative and Storytelling Gaps:** Many AR museum applications focus on visual spectacle without embedding strong narratives that help visitors build historical or scientific continuity. In geology, the absence of storytelling can cause visitors to miss the processes that connect fossils, climate shifts, and mass extinctions into a coherent Earth history.
- **Youth Engagement vs. Educational Depth:** Coloring-to-AR applications encourage creativity and personalization, but they risk being seen as “play-only” without structured integration into educational outcomes. For geological learning, the challenge is to design interactions that are both enjoyable and scientifically meaningful.
- **Technical and Usability Barriers:** QR-based, WebAR, and marker-based AR systems provide cost-effective access, yet they face issues such as browser compatibility, device capability, lighting sensitivity, camera quality, and scalability across large exhibit spaces. If not addressed, these limitations could restrict adoption or reduce the consistency of visitors' experiences.

Integration of Narration and Interaction: While AI-driven narration, WebAR access, and interactive AR features each enhance learning, combining them into a coherent visitor flow is challenging. Poorly synchronized narration, disjointed storytelling, or overlapping interactions can overwhelm users rather than guide them. For museum settings, ensuring that QR access, browser-based AR scenes, app-based AR scenes, voice narration, animations, coloring activities, and quizzes complement one another is crucial to achieving cultural and scientific learning outcomes.

By addressing challenges of storytelling integration, AR accessibility, skill-based engagement, usability constraints, and cultural interpretation, this study aims to contribute to a model for AR-enhanced museum experiences that are robust, trustworthy, and adaptable across educational and heritage domains.

This problem is not limited to the availability of technology. Even if AR and WebAR are available, the design must ensure that the visitor experience remains coherent, age-appropriate, and educationally useful. If visitors scan a QR code and only see an isolated 3D model, the experience may be attractive but not necessarily meaningful. The learning value depends on how the WebAR entry point, app-based AR scenes, narration, interaction, and assessment are organized into a connected sequence.

The broader Bridging Realities project attempts to address museum learning through multiple immersive components, including AR, WebAR, VR, AI narration, and interactive experiences. Within that larger project, this individual component focuses on how AR can support geological learning through a mobile-first and browser-accessible workflow that visitors can use with ordinary smartphones.

1.5 Research Objective

The main objective of this research is to design and implement an interactive Augmented Reality component that enhances how visitors learn about Earth's geological history within a museum environment. The study focuses on transforming static geological exhibits into immersive, story-driven, and interactive learning experiences by combining QR-triggered WebAR access, app-based AR visualization, AI-supported narration, Color-to-AR interaction, and quiz-based learning evaluation. Through this approach, the research aims to improve visitor engagement, support better understanding of geological eras, and provide an accessible digital learning experience suitable for children, students, and general museum visitors.

The specific objectives of this research are:

1. To develop QR-triggered WebAR and app-based AR storytelling experiences for geological eras.

This objective focuses on designing AR scenes for the Paleozoic, Mesozoic, and Cenozoic eras, where visitors can scan QR codes placed near exhibits to access WebAR scenes through a browser and app-based AR visualizations through the mobile component. These experiences will present key geological environments, prehistoric species, evolutionary changes, and major Earth history events in a more understandable and engaging manner.

2. To implement a Color-to-AR interaction for active visitor participation.

This objective aims to allow children and young visitors to color predefined geological illustrations and view their artwork as animated AR models. By connecting physical coloring activities with digital visualization, the system encourages creativity, interaction, and deeper involvement in the learning process.

3. To integrate AI-supported narration with AR-based visual learning.

This objective focuses on using AI-supported or pre-generated narration to guide visitors through AR scenes. The narration will explain geological concepts such as fossilization, mass extinctions, climate changes, plate movements, and the evolution of life using simple and age-appropriate language.

4. To evaluate the educational effectiveness and usability of the AR experience.

This objective focuses on assessing whether the proposed AR component improves visitor understanding and engagement. The evaluation will use pre-test and post-test quizzes, usability feedback, and visitor interaction data to measure learning improvement, ease of use, and overall user experience.

The first objective is important because geological eras are difficult to understand without visual and chronological support. By creating QR-triggered WebAR and app-based AR storytelling scenes, the system can divide complex Earth history into smaller learning moments. Each era-based scene is expected to present a set of related concepts, such as ancient environments, dominant species, and major transformations, in a sequence that visitors can follow.

The second objective supports engagement through active participation. Digital Color-to-AR interaction is included because children often learn better when they can create, manipulate, and observe outcomes rather than only listen or read. In this component, coloring is not treated as a separate art activity. Instead, it is connected to prehistoric or era-based visual content so that the user's creative input becomes part of the AR experience. This can increase attention, ownership, and curiosity while still supporting the museum's educational goal.

The third objective addresses explanation and accessibility. Geological concepts can be unfamiliar to many visitors, and static labels may not provide enough support. AI-supported or prepared narration can guide the visitor through the AR scene using simpler language, shorter explanations, and a clearer sequence. Narration also supports visitors who may prefer listening to reading, especially children and families moving through the gallery. In this way, the system combines visual, auditory, and interactive learning channels.

The fourth objective ensures that the project is evaluated as a learning tool, not only as a technical prototype. Pre-test and post-test quizzes are used to compare knowledge before and after the experience. Usability feedback and observation help identify whether users can scan markers, understand instructions, follow narration, complete coloring, and answer quizzes without confusion. These evaluation activities are necessary because museum technology can only be considered successful if it is both usable and educationally meaningful.

Together, the objectives support the main aim of the study by linking system development with measurable learning outcomes. The component is designed not only to demonstrate AR and WebAR functionality, but also to provide a repeatable model for interactive geological learning in museums. This objective structure ensures that the technical work, user interaction, narration, and evaluation are aligned with the research problem.

2. METHODOLOGY

2.1 Overview

The methodology of this research focuses on the design, development, and evaluation of an Augmented Reality based geological learning component for the Bridging Realities museum learning system. Its main purpose is to explain how the AR component was planned, structured, implemented, and tested to improve visitor engagement and understanding of geological concepts. The methodology follows a development-based approach, where interactive AR features, 3D geological content, QR-based access, color-to-AR activities, and knowledge-checking functions support child-friendly museum learning. This section also explains the project scope, system design, development workflow, tools, testing approach, and project timeline.

2.1.1 Research Design and Methodological Approach

This research adopts a mixed-methods design supported by an iterative development and evaluation process. The methodology is structured to examine both the technical feasibility of the AR and WebAR component and its educational effectiveness in a museum-based learning environment. Quantitative data is used to measure learning improvement, usability, and interaction behaviour, while qualitative feedback is used to understand visitor experience, clarity of content, and perceived engagement.

2.1.2 Individual Component Scope

The study is designed around the individual component, Interactive AR Exploration of Earth's Geological History, which focuses on presenting the Paleozoic, Mesozoic, and Cenozoic eras through mobile AR and QR-triggered WebAR access. The methodology therefore combines system development, WebAR entry design, digital content preparation, AR interaction design, and user evaluation. This allows the research to investigate not only whether the system can be implemented, but also whether the interaction model supports meaningful learning within the Colombo National Museum and Natural History Museum context.

2.1.3 Target Users and Learning Flow

The target users are museum visitors, students, and children who interact with geological exhibits. Participants experience the system using a smartphone or tablet. The user flow begins with a short pre-test to identify existing knowledge of geological eras, followed by QR scanning, WebAR or app-based AR interaction, narration, Color-to-AR activity, and post-test usability feedback. This approach supports comparison between the user's knowledge before and after the AR learning experience.

2.1.4 AR Interaction Modes

Two main interaction modes are included in the methodology. The first mode is QR-triggered geological era storytelling, where visitors scan a QR code or image marker linked to a specific era. The QR scan can open a WebAR scene through the browser for quick access, while the app-based AR route uses image target recognition to activate a richer narrated scene. In both routes, the system displays era-based 3D models, animations, labels, and contextual explanations.

The second mode is the digital Color-to-AR interaction, where users color a predefined digital drawing directly on a smartphone or tablet. The completed digital sketch is then used to update the appearance of the related AR model or scene, allowing the user to observe their own-colored version in an interactive augmented environment. This replaces the earlier physical paper-sketch workflow and makes the interaction more practical, reusable and suitable for mobile deployment.

2.1.5 Technical Development Approach

The technical implementation uses two complementary AR delivery layers. The first layer is the app-based AR component developed using Unity, C#, and Vuforia Engine for image target recognition, marker tracking, and AR content activation. The second layer is the WebAR QR access route, where selected geological content is opened through a mobile browser after scanning a QR code. WebAR supports quick access for visitors who may not install the application, while the Vuforia-based mobile component supports more controlled target tracking and interaction. Vuforia Image Targets are suitable for predefined QR-like visual markers, era cards, or exhibit boards because they detect and track known image features from the live camera image [17], [18].

The system development process follows an iterative prototyping approach. First, geological content is selected and organized into era-based narratives. Next, digital assets such as 3D models, UI screens, WebAR pages, narration scripts, quiz questions, and digital coloring templates are prepared. Then, the WebAR QR scenes and app-based AR scenes are implemented and connected with the relevant QR codes or image targets. After each prototype iteration, functional testing is conducted to check QR access, target recognition, scene loading, rendering performance, narration timing, quiz behaviour, and digital coloring output.

2.1.6 Data Collection and Analysis

Data collection is planned through pre-test and post-test quizzes, usability feedback, and observation of interaction behaviour. The quiz results are used to identify learning improvement, while usability feedback helps evaluate ease of use, clarity, and satisfaction. Observation is also used to understand whether visitors can scan a QR code, open the WebAR route, use the app-based AR route, follow narration, complete the Color-to-AR interaction, and answer the quiz without major support. The collected results are analysed using descriptive comparison of pre-test and post-test scores, supported by qualitative interpretation of visitor feedback.

2.1.7 Geological Content Organization

The methodology begins with content organization because the quality of the AR and WebAR experience depends on the accuracy and clarity of the geological story. The content is arranged around three major eras: Paleozoic, Mesozoic, and Cenozoic. For each era, the learning content is simplified into a short narrative, suitable 3D assets, narration points, interaction prompts, and quiz questions so that the same educational message can be delivered through both browser-based and app-based routes.

2.1.8 QR-Triggered AR Storytelling Design

The QR-triggered AR storytelling mode is implemented as an exhibit-linked interaction. Each QR code or image marker represents a specific geological era or learning point. When a QR code is scanned, the visitor can access the relevant WebAR content through a browser-based route. When an app-based image target is detected, the system activates the corresponding AR scene and displays the relevant 3D models and labels. In both cases, narration is used to guide the visitor through the scene rather than leaving the 3D content unexplained.

2.1.9 Digital Color-to-AR Interaction Design

The digital Color-to-AR workflow is designed to overcome practical limitations found in physical coloring systems. Physical paper coloring may require printing, paper management, camera scanning of user drawings, and stable lighting conditions. In a museum environment, these requirements can be difficult to maintain repeatedly. The digital coloring approach allows users to color directly on the phone or tablet, making the activity reusable and easier to control. The selected colors can then be reflected in the related AR model or scene, creating a direct connection between user input and augmented output.

2.1.10 AI-Supported Narration Design

The role of AI-supported narration in this methodology is to make the content easier to understand. The narration script is designed to avoid excessive scientific detail and instead explain each concept in a guided, age-appropriate manner. For example, instead of presenting mass extinction only as a scientific term, the narration can explain it as a major event where many living things disappeared and new life forms later developed. This style of explanation is more suitable for children and general museum visitors while still preserving scientific meaning.

2.1.11 Technology Selection

From a technical perspective, Vuforia Engine is selected for the app-based component because it provides stable image target recognition and marker-based triggering. Vuforia Image Targets are suitable for printed cards, QR-like visual markers, and era boards because the engine can detect and track known image targets using natural visual features [17], [18]. WebAR is used for the QR scan route because it allows selected AR content to be opened through a mobile browser, making the experience more accessible for visitors who do not want to install an application [13], [14].

2.1.12 Evaluation Design and Mixed-Methods Justification

The evaluation design follows a simple but structured pre-interaction and post-interaction comparison. Before using the system, participants complete a short pre-test to measure their existing understanding of geological eras. After interacting with the AR scenes and quiz module, their post-learning scores are reviewed to identify changes in knowledge. This approach is suitable for a prototype because it provides a measurable indication of learning improvement without requiring a long-term controlled study.

Qualitative feedback is used to support the numerical findings. While quiz scores show whether performance changed, they do not fully explain why users improved, struggled, or lost interest. Observation and usability feedback therefore help identify problems such as unclear instructions, difficulty scanning markers, confusing quiz wording, narration timing issues, or lack of visual clarity. These insights are important for improving the prototype before larger-scale museum testing.

The mixed-methods approach is appropriate because the research has both technical and educational goals. A purely technical evaluation would show whether the WebAR page opens or the app-based AR scene launches, but it would not show whether visitors understand the geological content. A purely educational evaluation would measure learning, but it would not identify issues such as QR access problems, marker recognition failures, browser limitations, narration timing, or mobile performance. Combining both perspectives gives a more complete understanding of the prototype.

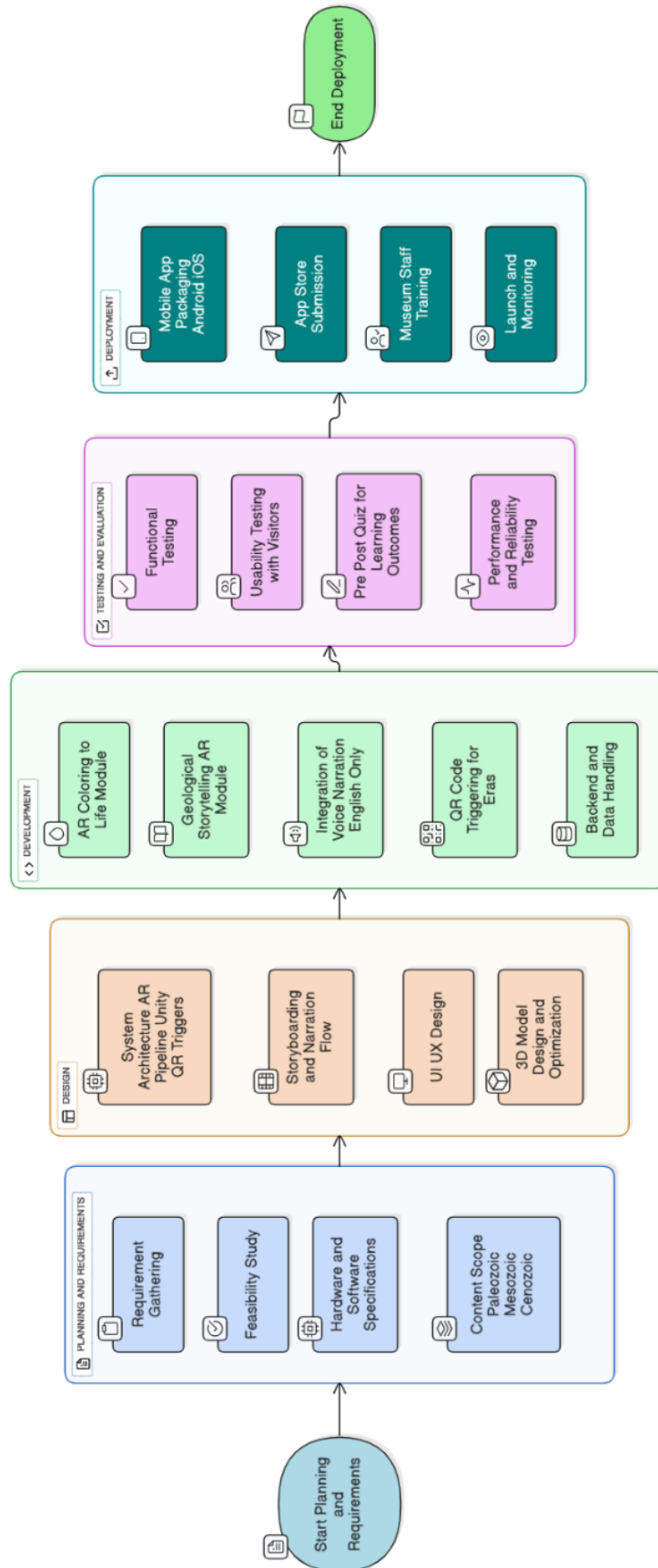


Figure 1: Work Breakdown Structure

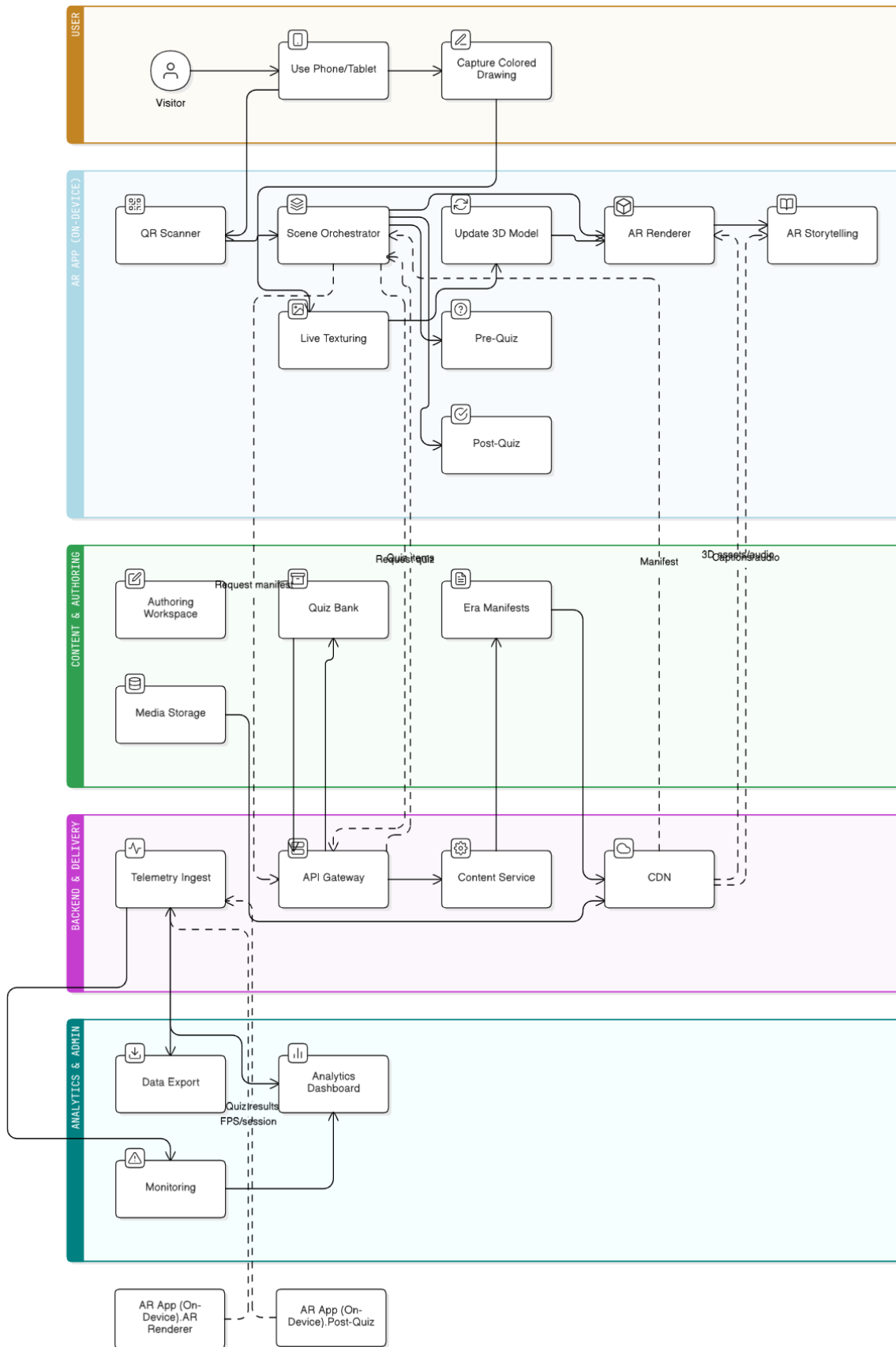


Figure 2: System Diagram of the Component

RESEARCH PROJECT PROCESS

GANTT CHART

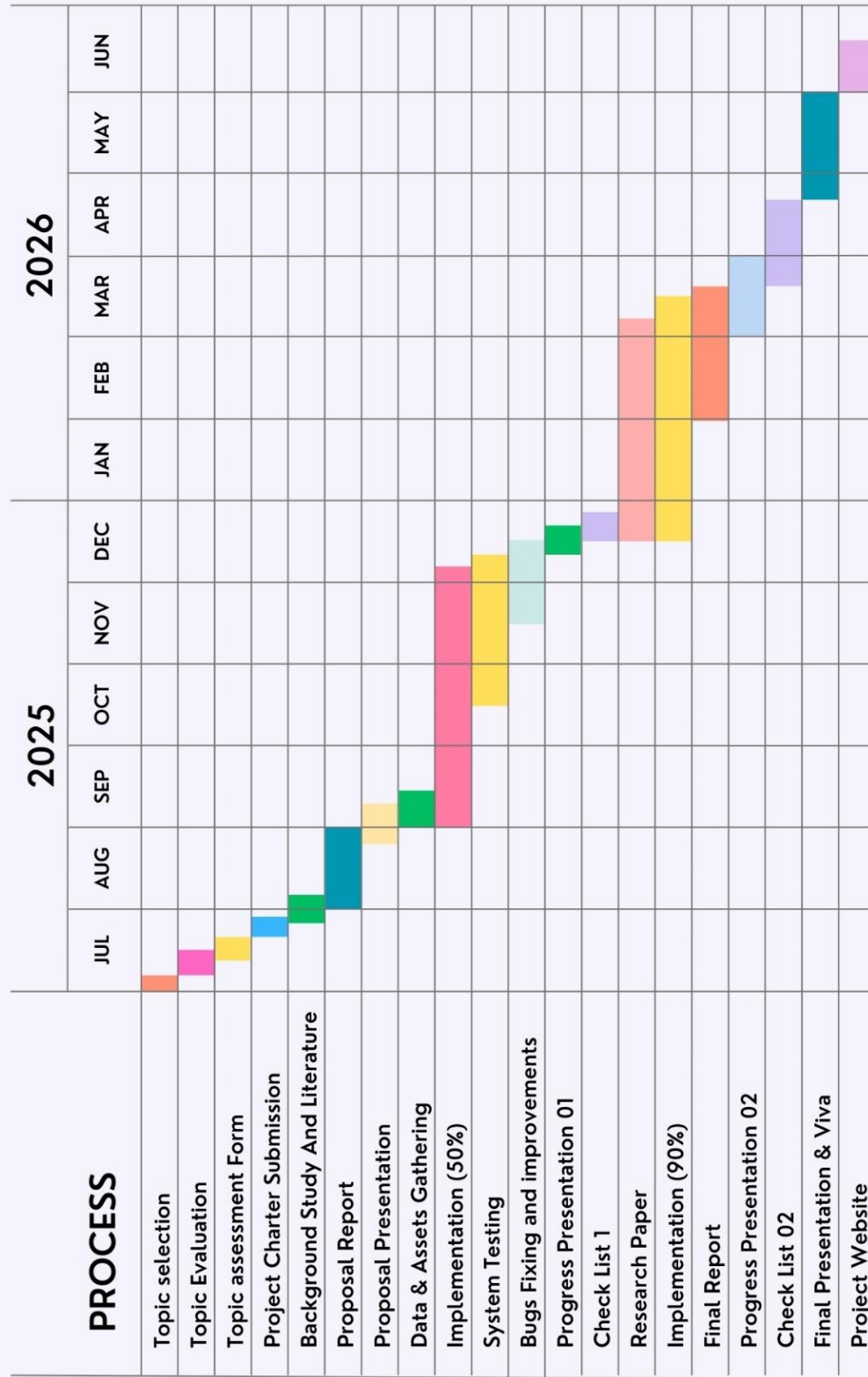


Figure 3: Gantt Chart

2.2 Commercialization Aspect of the Project

The proposed AR-based geological learning system has strong commercialization potential as an interactive digital learning solution for museums, schools, science exhibitions, and tourism-related educational environments. Museums in Sri Lanka and globally are increasingly looking for interactive and affordable ways to modernize static exhibits and attract younger audiences. Since geological history is difficult to explain through captions alone, the combination of QR-based WebAR access and app-based AR provides a marketable approach for transforming fossil, rock, and mineral displays into visual, interactive, and educational experiences.

The system can be commercialized as a mobile AR museum learning package that includes QR-triggered WebAR scenes, app-based 3D geological scenes, narration, quizzes, and Color-to-AR activities. Since it uses smartphones, printed markers, and browser-based access instead of expensive AR headsets, it is suitable for resource-constrained museums, schools, and science exhibitions.

2.2.1 Commercialization Options

The following commercialization options present possible ways in which the proposed system can be introduced, adapted, and expanded for different user groups and institutional needs. These options focus on practical use cases such as museum learning, school education, exhibitions, tourism support, multilingual accessibility, and feedback-based improvement.

1. **Museum Learning Package** : A ready-to-use AR learning package for fossil, rock, mineral, and geological era exhibits.
2. **QR-Based WebAR Exhibit Access** : QR codes placed near museum displays to allow visitors to access AR content through a browser.
3. **App-Based Guided AR Experience** : An enhanced AR mobile application for guided tours, school visits, workshops, and special exhibitions.
4. **School Science Learning Module** : An interactive learning module for teaching geological eras, fossilization, extinction, evolution, and Earth history.

5. **Science Exhibition and Event Package** : A portable AR experience for science fairs, educational exhibitions, STEM events, and mobile learning programs.
6. **Tourism-Focused Museum Experience** : An interactive and visually rich museum experience for both local and international visitors.
7. **Custom Content Development for Other Museums** : A customizable AR solution where different museums can add their own exhibits, models, narration, and quizzes.
8. **Multilingual Narration Extension** : An expanded version with Sinhala, Tamil, English, and other language support for wider accessibility.
9. **Curriculum-Aligned Educational Package** : A digital learning package aligned with school science syllabuses for teachers and students.
10. **Learning Analytics and Visitor Feedback Service** : A feedback-based service that helps museums understand visitor engagement and learning outcomes through quiz and interaction data.

These commercialization options show that the proposed system is not limited to a single museum environment. It can be adapted for different educational, cultural, and tourism-based contexts by modifying the content, language support, and learning activities. Therefore, the system has potential to become a flexible and scalable AR learning solution for improving museum education and visitor engagement.

2.2.2 Competitive Advantage

The main competitive advantage of this system is its low-cost and scalable structure. It combines geological storytelling, child-focused creativity, browser-accessible QR scanning, app-based AR visualization, AI-supported narration, Color-to-AR interaction, and quiz-based learning evaluation within one workflow. This makes it easier for museums to adopt the system gradually without redesigning the entire exhibition space.

Another strength is its modular design. Once the main workflow is developed, new geological eras, fossil types, local species, or museum exhibits can be added as separate content packages. This allows phased deployment and future expansion while keeping development and maintenance costs manageable.

2.3 Testing and Implementation

Testing and implementation are organized to verify the functional accuracy, technical stability, and educational suitability of the prototype. Since the system includes both WebAR QR access and app-based AR using Unity and Vuforia Engine, testing focuses on QR scanning, browser-based scene loading, image target recognition, AR scene activation, mobile rendering, digital sketch interaction, narration flow, and quiz-based learning evaluation.

The QR-triggered storytelling module is tested by scanning each era QR code and confirming whether the correct WebAR scene or app-based AR scene is launched. Testing also checks whether the 3D models, labels, animations, and narration appear in the correct sequence. The digital Color-to-AR module is tested by allowing users to color a predefined sketch using the mobile interface and then verifying whether the selected colors are reflected in the related AR model or scene. This confirms that the digital-sketch workflow works without requiring printed worksheets or physical scanning of colored paper.

The quiz module is tested using both pre-test and post-test question flows. This includes validating question display, answer selection, score calculation, and comparison of learning results. Usability testing focuses on whether children and general visitors can complete the core actions with minimal guidance: scan the QR code or marker, open WebAR or app-based AR content, view the scene, listen to narration, color the digital sketch, interact with the output, and answer quiz questions.

The original proposal-style project requirements, system requirements and risk considerations have been consolidated under this thesis-required section instead of remaining as separate standalone chapters. This keeps the document aligned with the final thesis structure while preserving the implementation details needed to explain the component.

Requirement Area	Summary
Functional	QR-triggered WebAR storytelling, app-based AR visualization, AI-supported narration, digital Color-to-AR interaction, pre/post quizzes, feedback and engagement tracking.
Non-functional	Fast loading, browser accessibility, child-friendly usability, captions, scalability, local/anonymized storage, maintainable Unity/WebAR content structure and offline/fallback support where possible.
System	Unity 3D, Vuforia Engine, C#, WebAR/WebXR-supported browser access, Blender/Meshy assets, digital sketch interface, narration audio, quiz database and Android mobile devices.
User	Visitors should be able to scan QR codes or markers, open WebAR or app-based AR content, view, listen, interact, digitally color, answer quizzes and understand era-based geological content with minimal guidance.

Table 2: System Requirements Summary

The system requirements summarized in *Table 2* provide the basis for planning the implementation and testing activities of the proposed AR component. Each requirement area is directly connected to a testable feature of the system, such as QR/WebAR access, Vuforia-based image target recognition, AR scene rendering, AI-supported narration, digital Color-to-AR interaction, and quiz-based learning evaluation. Therefore, the testing process was designed not only to confirm that the technical functions operate correctly, but also to verify whether the system supports the intended visitor learning flow. This ensures that the prototype is evaluated from both an implementation perspective and an educational usability perspective before being considered suitable for museum-based deployment.

Test Area	Test Focus	Success Criteria
QR / WebAR / marker scanning	Verify that each QR code opens the correct WebAR route and that Vuforia detects each app-based era target under normal museum lighting.	Pass if the correct WebAR or app-based AR scene launches consistently.
AR scene rendering	Check Paleozoic, Mesozoic and Cenozoic WebAR/app-based AR scene loading, model placement and animation.	Pass if scenes load smoothly without major lag or browser/app errors.
Digital Color-to-AR workflow	Color a predefined digital sketch on the phone/tablet and apply the output to the AR model/scene.	Pass if the selected colors are reflected in the relevant AR model or scene.
Narration synchronization	Check narration timing with visual scene events.	Pass if narration supports rather than interrupts the flow.
Quiz and scoring	Run pre/post quiz and score calculation.	Pass if answers and score are stored/calculated correctly.
Usability	Observe child/adult users completing QR scan, WebAR/app-based AR access, narration, coloring and quiz activities.	Pass if users can complete core tasks with limited support.

Table 3: Test Implementation Plan

2.3.1 Implementation testing

The implementation testing begins with functional verification of the main user flow. Each era QR code and image marker is tested separately to confirm that the correct WebAR or app-based AR scene is launched and that the scene remains stable while the mobile device is moved. This is necessary because unstable QR access, browser loading, or target recognition could interrupt the learning flow and reduce visitor confidence.

2.3.2 Narration testing

Narration testing focuses on timing and clarity. The audio or AI-supported narration must match the visual events shown in the AR scene. If the narration begins too early, ends too late, or explains content that is not visible on screen, users may become confused. Therefore, each scene is reviewed to ensure that narration, labels, and animation support one another. Captions or short text prompts are also important because museum environments may be noisy, and some users may prefer reading over listening.

2.3.3 Color-to-AR workflow testing

The digital Color-to-AR workflow is tested by checking whether the coloring interface is easy to understand and whether the selected colors are correctly applied to the intended AR output. This includes testing color selection, reset options, model update behavior, and transition from the coloring screen back to the AR view. Since this activity is aimed mainly at children, usability is especially important. The interaction should be simple enough to complete without long instructions.

2.3.4 Pre and post Quiz testing

Quiz testing verifies both accuracy and fairness. Each question must be linked to the content presented in the AR experience so that the quiz assesses learning from the system rather than unrelated prior knowledge. The pre-test and post-learning quiz flow must also be checked for score calculation, answer validation, and result display. Because the current prototype uses different era-based quizzes, the discussion section acknowledges that future evaluations should use more controlled equivalent tests for stronger measurement.

2.3.5 Performance testing

Performance testing is also considered because mobile AR and WebAR experiences can be affected by device capability, browser support, lighting, camera quality, network availability, and model complexity. The prototype is reviewed for loading time, frame stability, target detection, WebAR page response, and user interface responsiveness. If models are too detailed or scripts are too heavy, the experience may become slow on lower-end devices, so asset optimization is part of the testing process.

2.3.6 Usability testing

Usability testing observes whether users can complete the full experience without major support. This includes scanning the QR code or marker, opening the correct AR route, understanding the narration, interacting with the AR content, completing the coloring activity, and answering the quiz. The goal is to identify barriers that may prevent real museum visitors from completing the experience smoothly.

3. RESULTS AND DISCUSSION

3.1 Results

This section presents the prototype evaluation results of the Bridging Realities AR learning component. The evaluation focused on participant demographics, pre-quiz and post-learning performance, age-group-based learning differences, era-based quiz correctness, and overall learning outcomes. The purpose of this evaluation was to examine whether the proposed WebAR-supported and app-based AR geological learning experience could support short-term knowledge improvement and provide an engaging learning flow for museum visitors.

The results should be interpreted as prototype-level evaluation evidence. The dataset consists of participant profiles used to assess expected product behaviour and learning performance patterns. Therefore, the findings are useful for evaluating the design direction of the application, but they should not be treated as final evidence from a large-scale controlled museum study.

The evaluation dataset consisted of 27 participant profiles representing different age groups relevant to a museum learning context. The participant group included 5 children under the age of 13, 9 teenagers between 13 and 18, and 13 adults aged 18 and above. In terms of gender distribution, the dataset was relatively balanced, with 14 female participants and 13 male participants.

Variable	Category	Frequency	Percentage
Age Group	Kid (Under 13)	5	18.5%
Age Group	Teen (13-18)	9	33.3%
Age Group	Adult (18+)	13	48.1%
Gender	Female	14	51.9%
Gender	Male	13	48.1%

Table 4: Participant Demographics

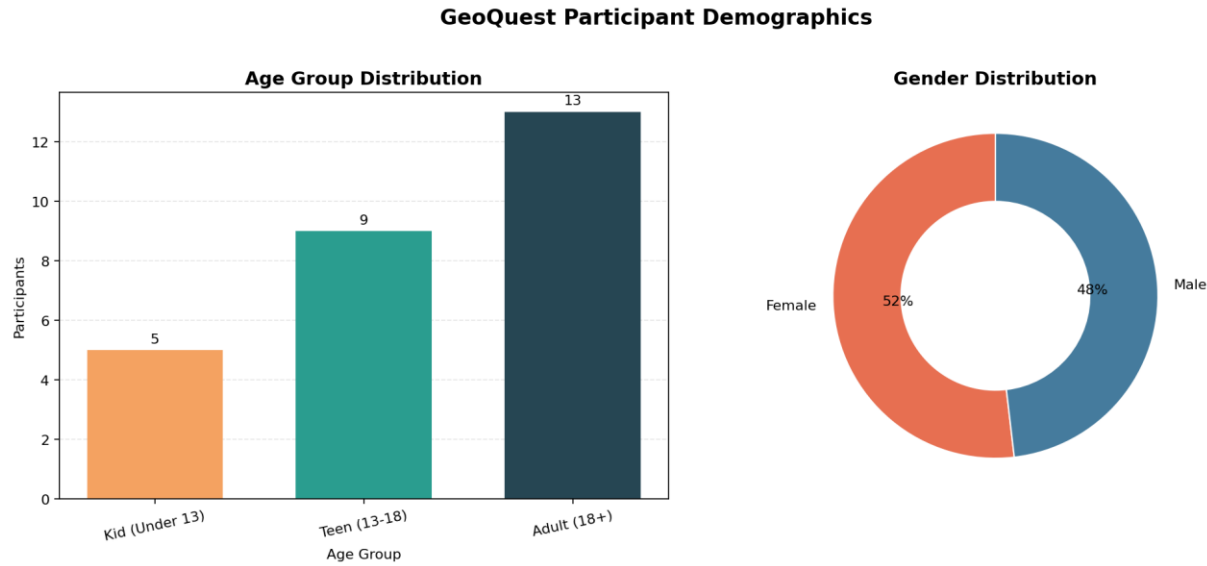


Figure 4: Participant demographics by age group and gender

To evaluate learning performance, participants' pre-quiz scores were compared against their weighted post-learning era quiz scores. The pre-quiz score represented the participant's baseline knowledge before interacting with the AR learning content. The weighted post-learning score was calculated using the participant's total correct answers across the attempted Paleozoic, Mesozoic, and Cenozoic quizzes, divided by the total number of questions attempted, and converted into a percentage.

Across all 27 participants, the average pre-quiz score was 58.02%, while the average weighted post-learning score increased to 62.98%. This represents an average improvement of 4.95 percentage points after interacting with the Bridging Realities learning experience.

Metric	Value
Total participants	27
Average pre-quiz score	58.02%
Average weighted post-learning score	62.98%
Average improvement	+4.95 percentage points
Improved users	16
No change	2
Declined users	9

Table 5: Overall Pre-Quiz and Post-Learning Performance

At the individual participant level, 16 out of 27 participants showed improvement after using the application, representing 59.3% of the dataset. Two participants showed no change, while nine participants recorded lower post-learning weighted scores compared to their pre-quiz baseline.

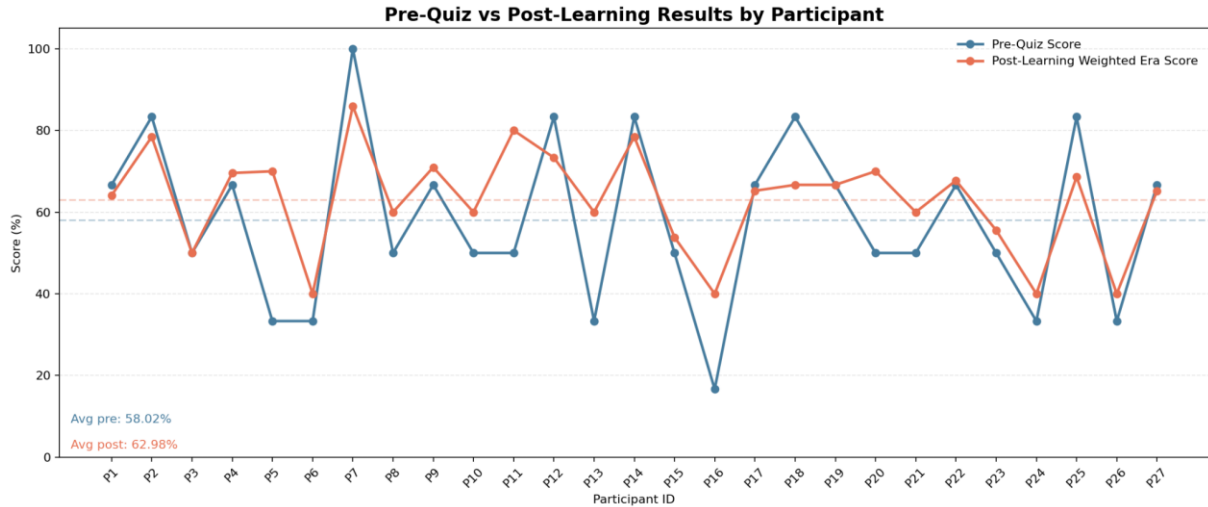


Figure 5: Pre-quiz vs post-learning results by participant

The age-group analysis provides further insight into how different user groups responded to the AR learning experience. Children recorded the largest average improvement, increasing from 43.33% in the pre-quiz to 51.33% in the post-learning weighted score. Teenagers improved from 57.41% to 64.15%, while adults showed a smaller increase from 64.10% to 66.64%.

Age Group	Participants	Average Pre-Quiz Score	Average Post-Learning Score	Average Change
Kid (Under 13)	5	43.33%	51.33%	+8.00 pp
Teen (13-18)	9	57.41%	64.15%	+6.74 pp
Adult (18+)	13	64.10%	66.64%	+2.54 pp

Table 6: Learning Performance by Age Group

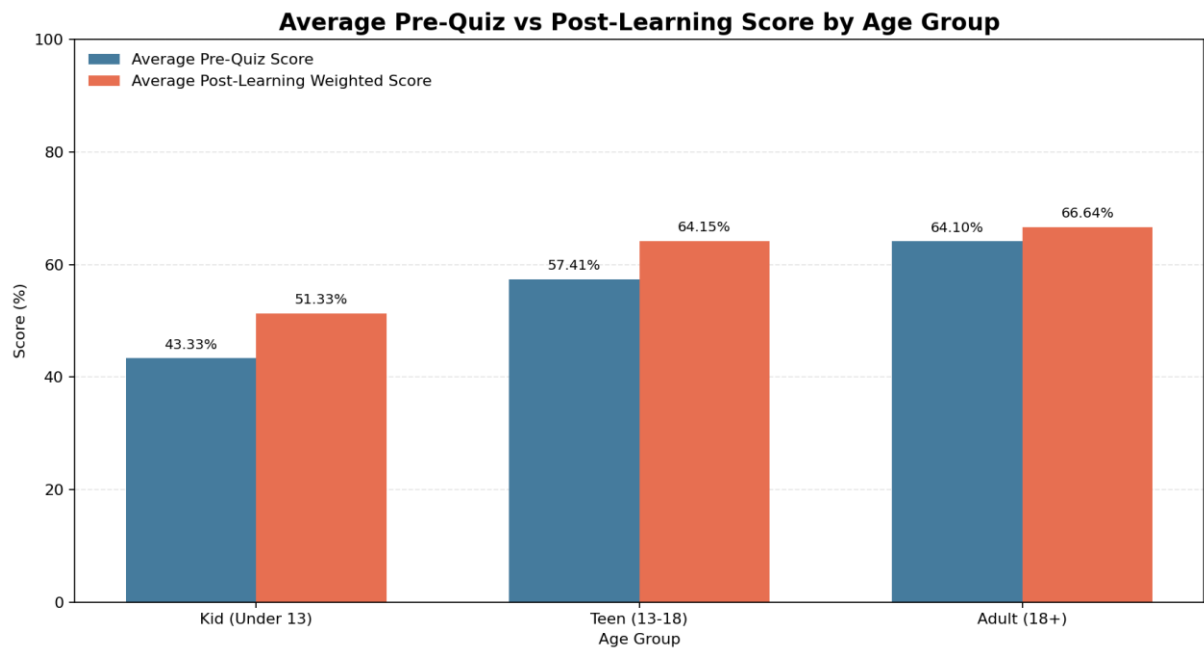


Figure 6: Average pre-quiz vs post-learning score by age group

The distribution of learning outcomes further supports the observation that the application had a positive learning effect for the majority of participants. Out of the 27 participants, 16 improved, 2 showed no change, and 9 declined in their post-learning weighted score.

Learning Outcome	Participants	Percentage
Improved	16	59.3%
No Change	2	7.4%
Declined	9	33.3%

Table 7: Distribution of Learning Outcomes

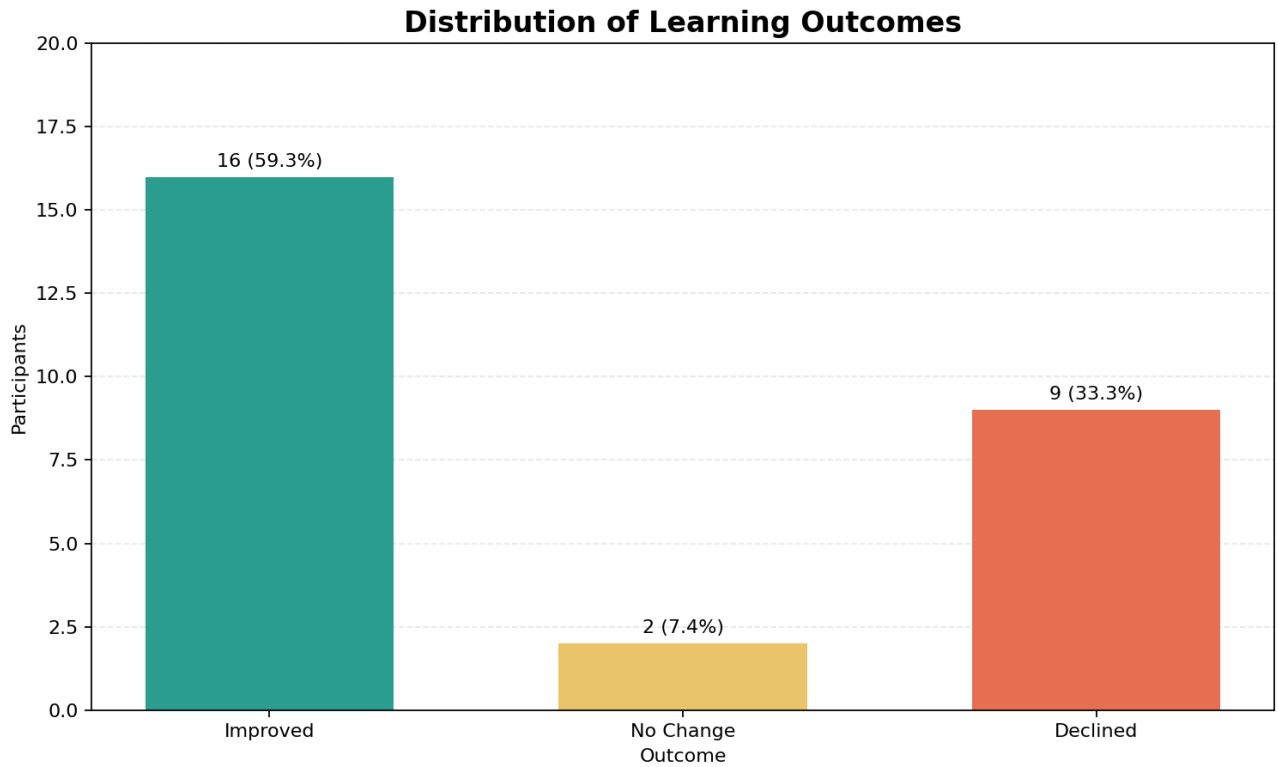


Figure 7: Distribution of learning outcomes

A quiz-level correctness summary was also examined to understand whether performance varied by era and difficulty level. The easy-level quizzes showed stronger average correctness across all three eras, while hard-level quizzes recorded lower averages. This suggests that the question difficulty design worked as intended, but it also means the weighted post-learning score must be interpreted carefully when participants attempted different combinations of easy, medium, and hard questions.

Quiz Area	User Score	Average Correct	Total Questions	Average Percentage
Pre-Quiz	27	3.48	6	58.02%
Paleozoic Easy	24	3.83	5	76.67%
Paleozoic Medium	14	5.36	8	66.96%
Paleozoic Hard	5	2.80	6	46.67%
Mesozoic Easy	27	3.52	5	70.37%
Mesozoic Medium	10	5.90	8	73.75%
Mesozoic Hard	6	2.50	6	41.67%
Cenozoic Easy	13	3.69	5	73.85%
Cenozoic Medium	7	5.43	8	67.86%
Cenozoic Hard	1	3.00	6	50.00%

Table 8: Era-Based Quiz Correctness Summary

Overall, the results show a moderate positive improvement in learning performance after participants interacted with the Bridging Realities AR learning experience. The improvement was especially visible among younger participants, suggesting that the visual, interactive, scan-based, and quiz-based structure of the application is more effective for children and teenagers.

The participant distribution provides useful context for interpreting the results. Since adults represented the largest group, the overall average is partly influenced by users who may already have more general knowledge and stronger reading ability. Children and teenagers, however, are highly relevant to the project because the AR component is designed to support younger museum visitors. Therefore, the age-group comparison is important for understanding whether the visual and interactive structure of the system is effective for the intended educational audience.

The average improvement of 4.95 percentage points should be interpreted as an early indication rather than a conclusion. In informal museum learning, even a moderate increase can be meaningful because visitors interact with the content for a limited time and without formal classroom instruction. The improvement suggests that the AR experience may help users recall or understand some geological concepts more effectively after interaction. However, the results also show that the prototype needs refinement because not all users have improved.

The strong improvement among children and teenagers supports the design decision to include visual storytelling, animated content, and interactive activities. Younger visitors often benefit from concrete representations of abstract concepts. When an era is shown through a scene, model, narration, and quiz, the content becomes easier to connect with memory. This finding is particularly important for the Colombo National Museum context, where school groups and family visitors are likely to form an important user segment.

The era-based quiz summary also provides useful design feedback. Easy-level questions achieved higher correctness, while hard-level questions produced lower scores. This pattern is expected and indicates that the difficulty levels can separate basic recognition from deeper understanding. At the same time, it reveals the need for careful quiz balancing. If participants attempt different difficulty levels, their scores may not be directly comparable. Future testing should therefore include a standardized post-test alongside optional era quizzes.

3.2 Research Findings

- The first major finding is that the Bridging Realities AR prototype demonstrated a positive short-term learning effect. The average score increased from 58.02% in the pre-quiz to 62.98% in the weighted post-learning era quizzes. This indicates that interaction with the WebAR-supported and app-based AR geological learning content was associated with improved quiz performance.
- The second finding is that the improvement in learning was stronger among younger users. Children showed an average improvement of 8.00 percentage points, while teenagers improved by 6.74 percentage points. Adults showed a smaller improvement of 2.54 percentage points. This suggests that the interactive and visual nature of the application is particularly suitable for school-age visitors and family-based museum audiences.
- The third finding is that most participants showed measurable improvement after using the application. Since 59.3% of participants improved compared to their pre-quiz baseline, the results suggest that the Bridging Realities experience has potential as a supportive educational tool for museum-based learning. However, the presence of 9 participants who scored lower after the learning experience shows that the application requires further refinement in content delivery, quiz balancing, and user guidance.
- The fourth finding is that the post-learning performance should be interpreted carefully because the weighted post-learning score combines different era-based quizzes. Since participants may have attempted quizzes with different levels of difficulty, a direct comparison with the pre-quiz score may not fully represent knowledge growth. Therefore, while the results are promising, they should be treated as prototype-level evidence rather than final proof of educational effectiveness.

Finding	Evidence	Interpretation
Positive overall learning gain	Pre: 58.02%, Post: 62.98%, Change: +4.95 pp	The AR experience may support short-term knowledge improvement.
Stronger effect for younger users	Kids: +8.00 pp, Teens: +6.74 pp	The application is especially suitable for children and teenagers.
Majority of users improved	16 out of 27 users improved	Most participants benefited from the learning experience.
Some users declined	9 out of 27 users declined	Usability, quiz difficulty, or content clarity may need improvement.
Age-related learning pattern observed	Adults had the highest baseline but smallest gain	Adults may have had less room for improvement due to higher prior knowledge.

Table 9: Summary of Research Findings

Further findings are that the prototype appears more effective as a guided learning supplement than as a standalone replacement for museum interpretation. The AR and WebAR experience can attract attention and explain selected concepts, but it works best when the content remains connected to the physical exhibit, narration, and teacher or curator guidance when available. The findings also show the importance of matching interaction design with user age. Children may respond positively to coloring, animation, and short narration, while adults may prefer more detailed explanations or optional deeper information. This means future versions should consider flexible content layers, where the basic story remains simple but additional information can be accessed by users who want more depth. Such a structure would make the system more inclusive for mixed museum audiences.

3.3 Discussion

3.3.1 Educational Impact of the AR Prototype

The results indicate that the Bridging Realities AR prototype can support short-term learning improvement in a museum-oriented educational environment. The increase from 58.02% to 62.98% suggests that the combination of AR visualization, era-based storytelling, digital interaction, and quiz-based learning has the potential to improve visitor understanding of geological history. Although the average improvement of 4.95 percentage points is moderate, it is meaningful for a prototype-stage system designed for informal learning rather than formal classroom instruction.

The moderate overall improvement can be explained by the nature of the learning experience. The system is designed for short museum interactions rather than extended classroom teaching. Visitors may only spend a few minutes with each AR scene, so the goal is to create immediate understanding and curiosity. In that context, the observed increase suggests that AR storytelling can help visitors form clearer impressions of geological eras.

3.3.2 Interpretation of Age-Group Differences

A key observation is that younger participants showed greater learning improvement than adults. Children and teenagers benefited more from the application, which may be due to the visual and interactive design of the AR experience. Features such as era-based exploration, digital coloring, 3D visualization, and quiz feedback may have helped younger users engage more actively with the learning content.

Adults showed a smaller improvement, increasing by only 2.54 percentage points. This does not necessarily indicate that the application was ineffective for adults. Instead, adults had the highest pre-quiz baseline score at 64.10%, which means they may already have had better prior knowledge compared to children and teenagers. As a result, there was less room for measurable improvement, suggesting a possible ceiling effect among adult participants.

3.3.3 Interpretation of Declined Scores

The decline observed in 9 participants should be interpreted carefully. A lower post-learning weighted score does not always mean that learning did not occur. Since the post-learning score was calculated from attempted era-based quizzes, participants who attempted more difficult questions may have received lower percentages despite engaging more deeply with the content. Future evaluations should therefore use a more controlled post-test structure with equivalent difficulty levels across all participants.

3.3.4 Museum Learning and Engagement Value

The age-group findings are important for the overall research objective because the project aims to make geological history more understandable and engaging for museum visitors, especially children and young learners. The stronger gains among children and teenagers support the suitability of AR-based storytelling and interaction for youth-focused museum education. This aligns with the project's goal of transforming static geological exhibits into participatory learning experiences.

The results also suggest that the application has potential as a digital companion for museum visits. By combining QR-triggered access, app-based AR scenes, AI-supported narration, digital Color-to-AR interaction, and quizzes, the system can guide users through geological eras in a structured way. This is particularly valuable for the Paleozoic, Mesozoic, and Cenozoic eras, which are difficult to communicate through static fossils, rocks, and text labels alone.

3.3.5 Technical Deployment Discussion

From an implementation perspective, the combined use of browser-based QR access and Vuforia Engine supports the project's museum deployment goal. QR access allows selected content to be opened quickly through a browser, reducing the need for immediate app installation. The Vuforia-based app route supports more controlled image or QR-like target recognition for richer AR scenes. Together, these routes allow the system to be integrated into existing museum exhibits without expensive hardware or major physical changes to the gallery.

The use of marker recognition and QR access also introduces design responsibilities. Markers and QR codes must have clear visual features, adequate size, and good placement in the gallery. Lighting conditions, browser support, network availability, and camera permissions must be considered, and the AR scenes should provide fallback instructions when scanning or loading fails.

3.3.6 Role of Narrative, Color-to-AR, and Narration

The results emphasize the importance of narrative structure. Geological history is not simply a collection of facts; it is a sequence of transformations involving life, climate, landforms, extinction, and adaptation. Static displays often present specimens separately, which can make the story difficult to follow. The AR component helps by organizing content into era-based scenes, supporting the idea that museum technology should not only show objects but also explain relationships between objects, events, and time periods.

The Color-to-AR feature has value for child engagement because it changes the visitor from an observer into a contributor. When users color a digital illustration and see the result reflected in AR, the interaction creates a stronger personal connection to the content. However, the activity must remain linked to learning goals so that it supports geological understanding rather than becoming only an entertainment feature.

AI-supported narration also requires careful design. While AI can make explanations more flexible and accessible, museum content must remain accurate, concise, and appropriate for children. For this reason, narration should be guided by approved scripts, controlled prompts, or reviewed content boundaries. In future development, multilingual narration in Sinhala, Tamil, and English would further improve accessibility for local and international visitors.

3.3.7 Limitations of the Current Evaluation

Several limitations must be acknowledged. First, the evaluation dataset is relatively small, with only 27 participants. Second, the number of children is limited, even though children are one of the primary target groups. Third, the weighted post-learning score combines different era-based quizzes, which may introduce difficulty variation across participants. These limitations mean that the findings should be interpreted as prototype-level evidence rather than final controlled proof of educational effectiveness.

Museum learning is influenced by prior knowledge, reading ability, attention span, device familiarity, and the amount of time spent in each activity. Therefore, the current pre-quiz and post-learning comparison provides an early indication of knowledge improvement but should be strengthened through equivalent question sets, balanced participant groups, and delayed retention testing.

3.3.8 Future Evaluation Improvements

Future evaluation should involve a larger sample of real museum visitors, especially children and school groups. Additional measures such as time spent in each AR scene, number of scans, narration completion, quiz retry attempts, coloring activity duration, and user satisfaction ratings would provide a stronger understanding of both engagement and learning effectiveness.

The design should also be tested in a real gallery environment. Museum spaces can include uneven lighting, visitors' movement, background noise, limited standing space near exhibits, and variable mobile connectivity. These factors can affect QR scanning, scene loading, marker recognition, narration clarity, and user attention. Therefore, future implementation should include on-site testing under realistic museum conditions.

Content governance is also important for a museum learning system. Since the component explains scientific and historical information, the narration scripts, quiz questions, and 3D representations should be reviewed by subject experts before wider deployment. This is especially important when simplifying complex geological ideas for children. The explanation must be easy to understand without becoming scientifically inaccurate.

3.3.9 Summary of Discussion

In summary, the results provide encouraging evidence that the Bridging Realities AR prototype can improve geological learning outcomes, especially among younger users. The findings show that AR-based 3D visuals, narration, quizzes, and interactive activities helped make geological concepts more engaging and easier to understand than static museum displays.

The results also support the research objective of transforming passive museum exhibits into interactive educational experiences. However, further improvements are needed through larger-scale testing, better usability refinement, multilingual narration, more controlled evaluation methods, and subject-expert review. With these improvements, the system has strong potential to support digital museum learning in Sri Lankan natural history contexts.

4. SUMMARY OF EACH STUDENT'S CONTRIBUTION

The overall research project consists of four interconnected components under the Bridging Realities theme. Each component contributes to improving museum storytelling and interaction through immersive technologies. The individual contribution focuses on the Interactive AR Exploration of Earth's Geological History component.

Student	Component / Contribution
Tennakoon A S B	Fossil Frontier VR - Virtual Fossil Excavation Adventure
Pinto M J I	Interactive AR Mobile Guide with RFID and AI Voice Narration for Fossil Exploration
Pradeepa W D R	Interactive AR Exploration of Earth's Geological History: QR-triggered WebAR access, app-based AR era storytelling, Color-to-AR interaction, AR scene design, narration flow and quiz-based evaluation.
Jieshe S L I	Design and Evaluation of a Multiplayer VR Museum with Linked Jungle Simulation and AI Voice Guidance

Table 10: Group Member Contributions

My individual contribution includes reviewing AR, WebAR, and Color-to-AR literature, identifying the research gap for geological storytelling, designing the interaction flow for Paleozoic, Mesozoic, and Cenozoic era exploration, preparing the system workflow, defining evaluation instruments, and planning implementation/testing activities for the AR component.

5. CONCLUSION

This individual research component proposes an AR-based and WebAR-supported approach to improve how visitors understand Earth's geological history in a museum environment. By combining QR-triggered WebAR access, app-based AR storytelling, 3D visualization, AI-supported narration, Color-to-AR interaction, and quiz-based assessment, the component provides a practical method for transforming passive fossil and rock displays into interactive learning experiences.

The study shows that a mobile-first and browser-accessible approach is suitable for museum contexts because it reduces dependency on expensive hardware and can be integrated with existing exhibits through QR codes and image-based triggers. The proposed workflow allows visitors to scan, view, listen, interact, and answer quizzes in a guided sequence, helping them understand the Paleozoic, Mesozoic, and Cenozoic eras more clearly.

The prototype evaluation indicates a positive short-term learning effect, especially among children and teenagers. This suggests that visual storytelling, digital coloring, narration, and quiz feedback can support engagement and improve understanding when they are connected to clear educational objectives. However, the current study remains prototype-level, and future testing should include a larger sample of real museum visitors with more standardized pre-test and post-test questions.

Future improvements should focus on multilingual narration, additional geological scenes, stronger AR compatibility, improved Color-to-AR accuracy, better analytics, and larger-scale visitor evaluation. Overall, this component contributes to the broader Bridging Realities project by demonstrating how AR and WebAR can support accessible, story-driven, and measurable geological learning within the Sri Lankan museum context.

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7. GLOSSARY

Term	Description
Augmented Reality	A technology that overlays digital content such as 3D models, text, audio or animations onto the real-world view.
Color-to-AR	An interaction where a user colors a digital drawing on a phone or tablet and views the colored result as an AR model or animation.
Geological Era	A major division of geological time used to describe Earth's history and life evolution.
Image Target	A 2D image or marker that can be detected by an AR system to trigger digital content.
System Usability Scale	A standardized questionnaire used to evaluate perceived usability.
WebAR	A browser-based augmented reality approach where users can access AR content through a web page, often after scanning a QR code, without immediately installing a separate mobile application.
QR-triggered AR	An interaction method where scanning a QR code opens or activates related augmented reality content linked to a specific exhibit or learning topic.

8. APPENDIX

8.1 Appendix A: Screenshots of Application Interfaces

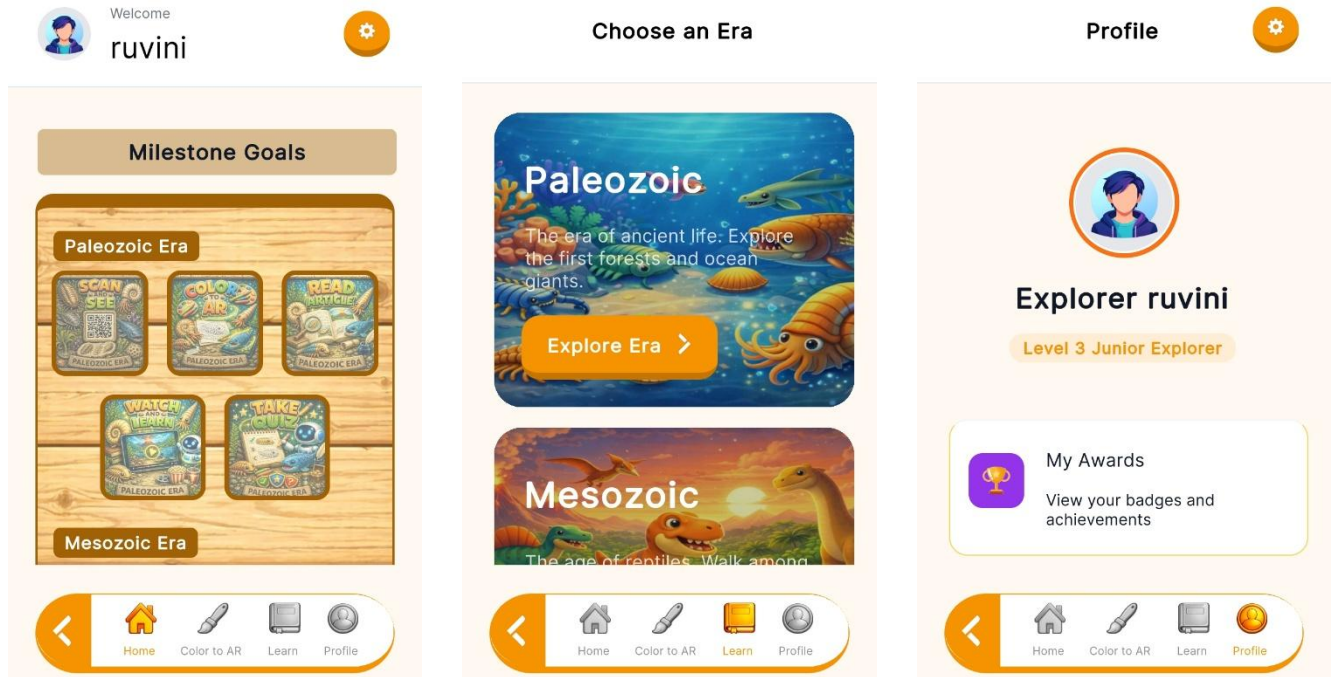


Figure 8: Main menu and profile interface screens

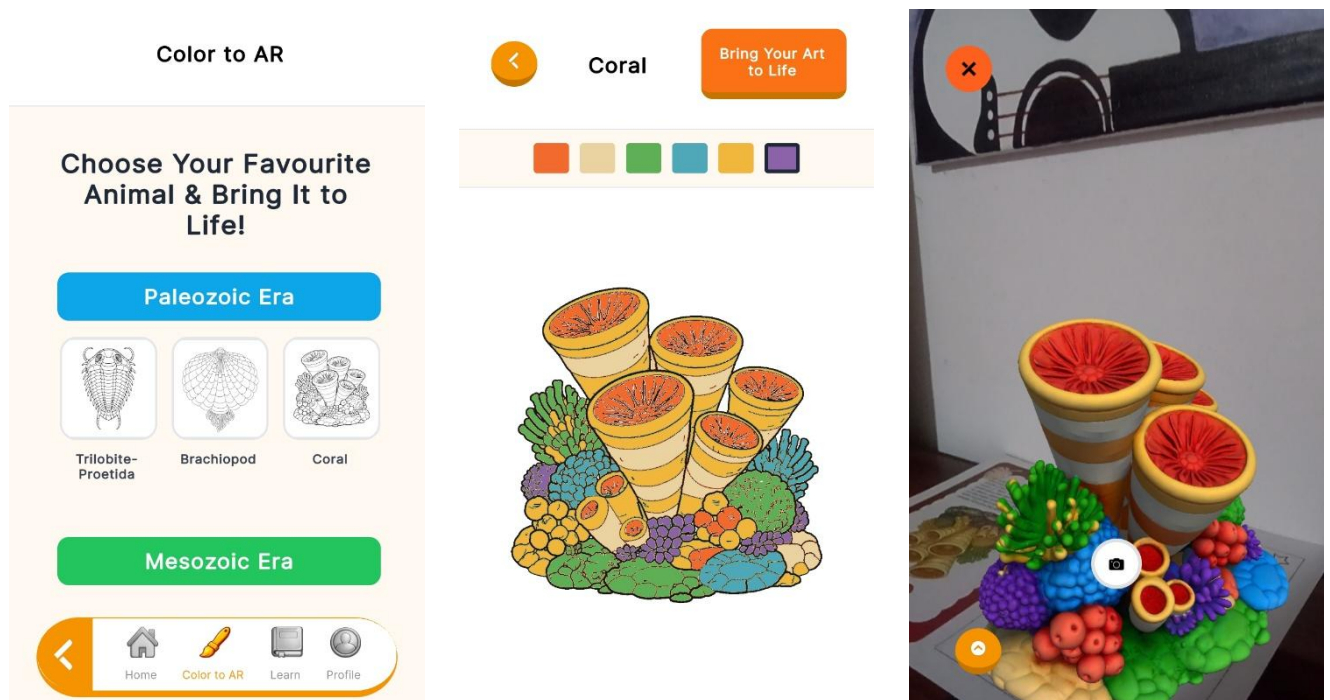


Figure 9: Color-to-AR selection, coloring, and AR output screens

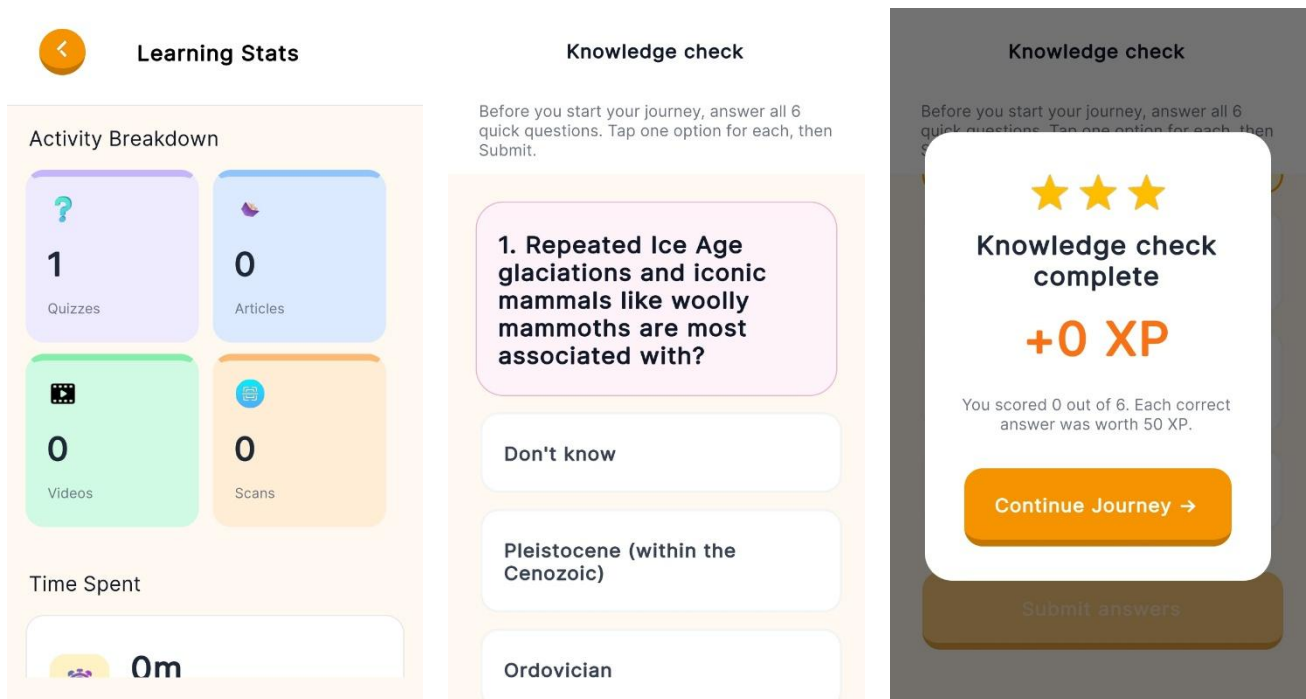


Figure 11: Learning statistics and knowledge check interface screens

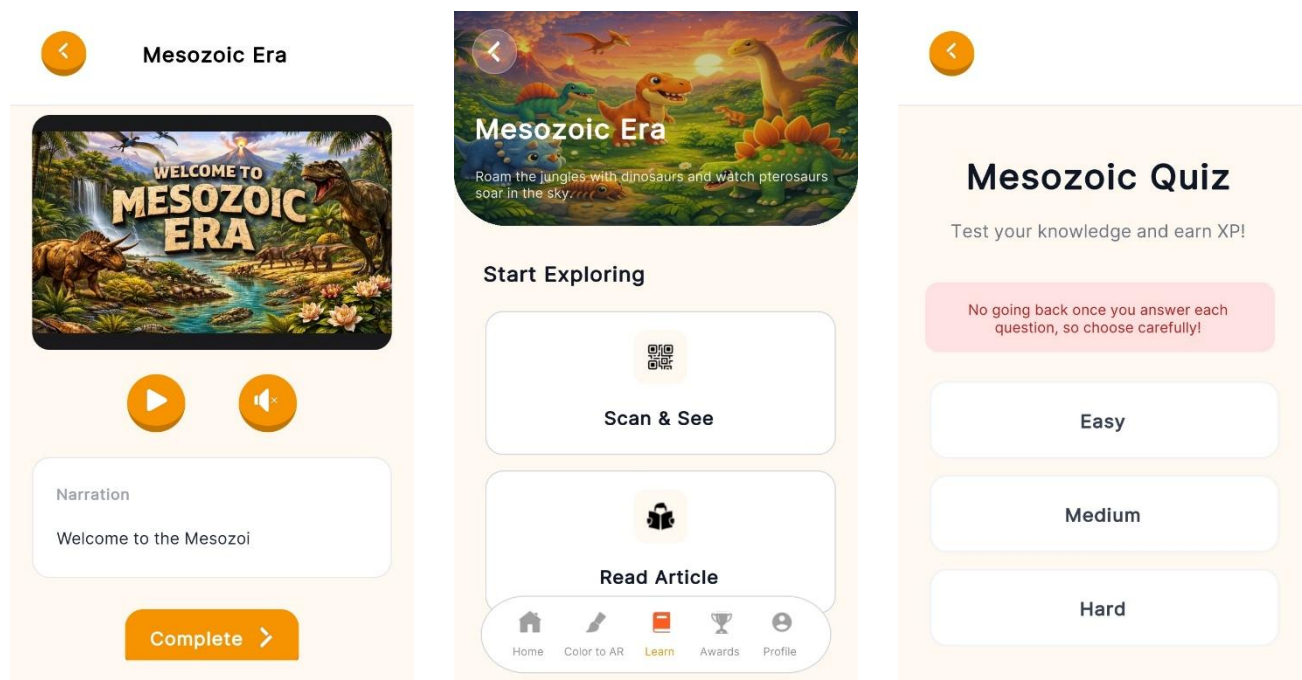


Figure 10: Era-based learning screens